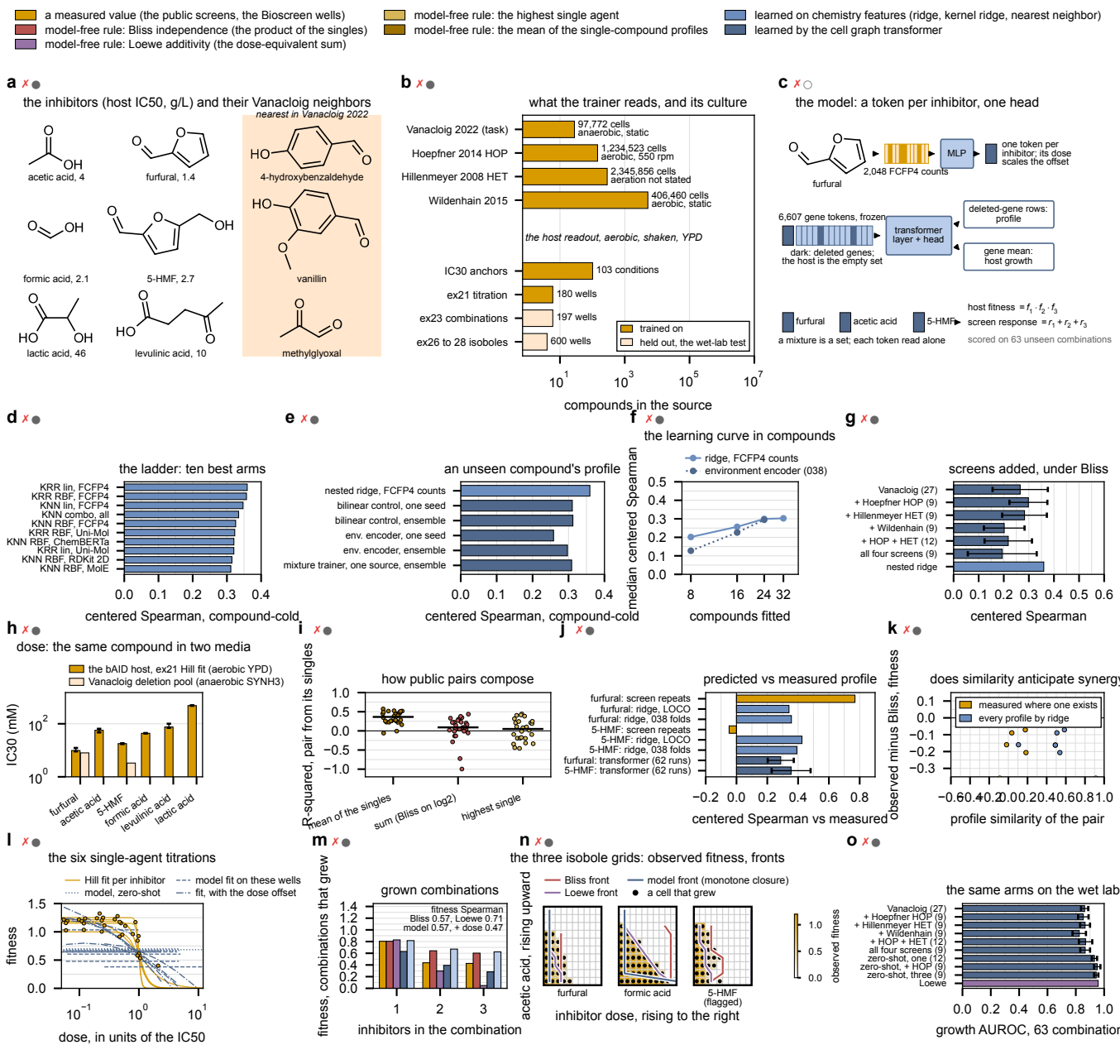

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the inhibitors (host IC50, g/L) and their Vanacloig neighbors
nearest in Vanacloig 2022
acetic acid, 4
furfural, 1.4
4-hydroxybenzaldehyde
formic acid, 2.1
5-HMF, 2.7
vanillin
lactic acid, 46
levulinic acid, 10
methylglyoxal
b
what the trainer reads, and its culture
Vanacloig 2022 (task)
97,772 cells
anaerobic, static
Hoepfner 2014 HOP
1,234,523 cells
aerobic, 550 rpm
Hillenmeyer 2008 HET
2,345,856 cells
aeration not stated
Wildenhain 2015
406,460 cells
aerobic, static
the host readout, aerobic, shaken, YPD
IC30 anchors
103 conditions
ex21 titration
180 wells
ex23 combinations
197 wells
ex26 to 28 isoboles
600 wells
trained on
held out, the wet-lab test
compounds in the source
c
the model: a token per inhibitor, one head
furfural
2,048 FCFP4 counts
MLP
one token per inhibitor; its dose scales the offset
6,607 gene tokens, frozen
dark: deleted genes; the host is the empty set
transformer layer + head
deleted-gene rows: profile
gene mean: host growth
furfural
acetic acid
5-HMF
a mixture is a set; each token read alone
host fitness = $f_1 \cdot f_2 \cdot f_3$
screen response = $r_1 + r_2 + r_3$
scored on 63 unseen combinations
d
the ladder: ten best arms
KRR lin, FCFP4
KRR RBF, FCFP4
KNN lin, FCFP4
KNN combo, all
KNN RBF, FCFP4
KRR RBF, Uni-Mol
KNN RBF, ChemBERTa
KRR lin, Uni-Mol
KNN RBF, RDKit 2D
KNN RBF, MolE
centered Spearman, compound-cold
e
an unseen compound's profile
nested ridge, FCFP4 counts
bilinear control, one seed
bilinear control, ensemble
env. encoder, one seed
env. encoder, ensemble
mixture trainer, one source, ensemble
centered Spearman, compound-cold
f
the learning curve in compounds
ridge, FCFP4 counts
environment encoder (038)
median centered Spearman
compounds fitted
g
screens added, under Bliss
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Wildenhain (9)
+ HOP + HET (12)
all four screens (9)
nested ridge
centered Spearman
h
dose: the same compound in two media
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
i
how public pairs compose
R-squared, pair from its singles
mean of the singles
sum (Bliss on log2)
highest single
j
predicted vs measured profile
furfural: screen repeats
furfural: ridge, LOCO
furfural: ridge, 038 folds
5-HMF: screen repeats
5-HMF: ridge, LOCO
5-HMF: ridge, 038 folds
furfural: transformer (62 runs)
5-HMF: transformer (62 runs)
centered Spearman vs measured
k
does similarity anticipate synergy
measured where one exists
every profile by ridge
observed minus Bliss, fitness
profile similarity of the pair
l
the six single-agent titrations
Hill fit per inhibitor
model, zero-shot
model fit on these wells
fit, with the dose offset
fitness
dose, in units of the IC50
m
grown combinations
fitness Spearman
Bliss 0.57, Loewe 0.71
model 0.57, + dose 0.47
fitness, combinations that grew
inhibitors in the combination
n
the three isobole grids: observed fitness, fronts
Bliss front
Loewe front
model front (monotone closure)
a cell that grew
acetic acid, rising upward
furfural
formic acid
5-HMF (flagged)
inhibitor dose, rising to the right
observed fitness
o
the same arms on the wet lab
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Wildenhain (9)
+ HOP + HET (12)
all four screens (9)
zero-shot, one (12)
zero-shot, + HOP (9)
zero-shot, three (9)
Loewe
growth AUROC, 63 combinations